

General Market Impact Classifier

Solution Architecture & Workflow

1. The Problem

Financial markets react to political events in real time. A single tweet from a world leader can move gold, equities, oil, and currencies within minutes — but most of those tweets are noise. The challenge is:

> **Given a political tweet and its timestamp, determine:**

- > 1. Is this tweet market-relevant at all?
- > 2. If yes — in which direction does it move each of seven financial assets at the 1m, 5m, and 10m horizons?
- > 3. Why — what specific evidence from the tweet and current market context drives that conclusion?

Manual analysis is too slow. A simple keyword filter misses nuance. A single model cannot capture entities, sentiment, event type, graph propagation, historical patterns, and live market context simultaneously.

The solution: a seven-layer pipeline that decomposes this problem into specialized stages, each contributing a distinct type of signal, and synthesizes them into a final verdict using an LLM with live market grounding.

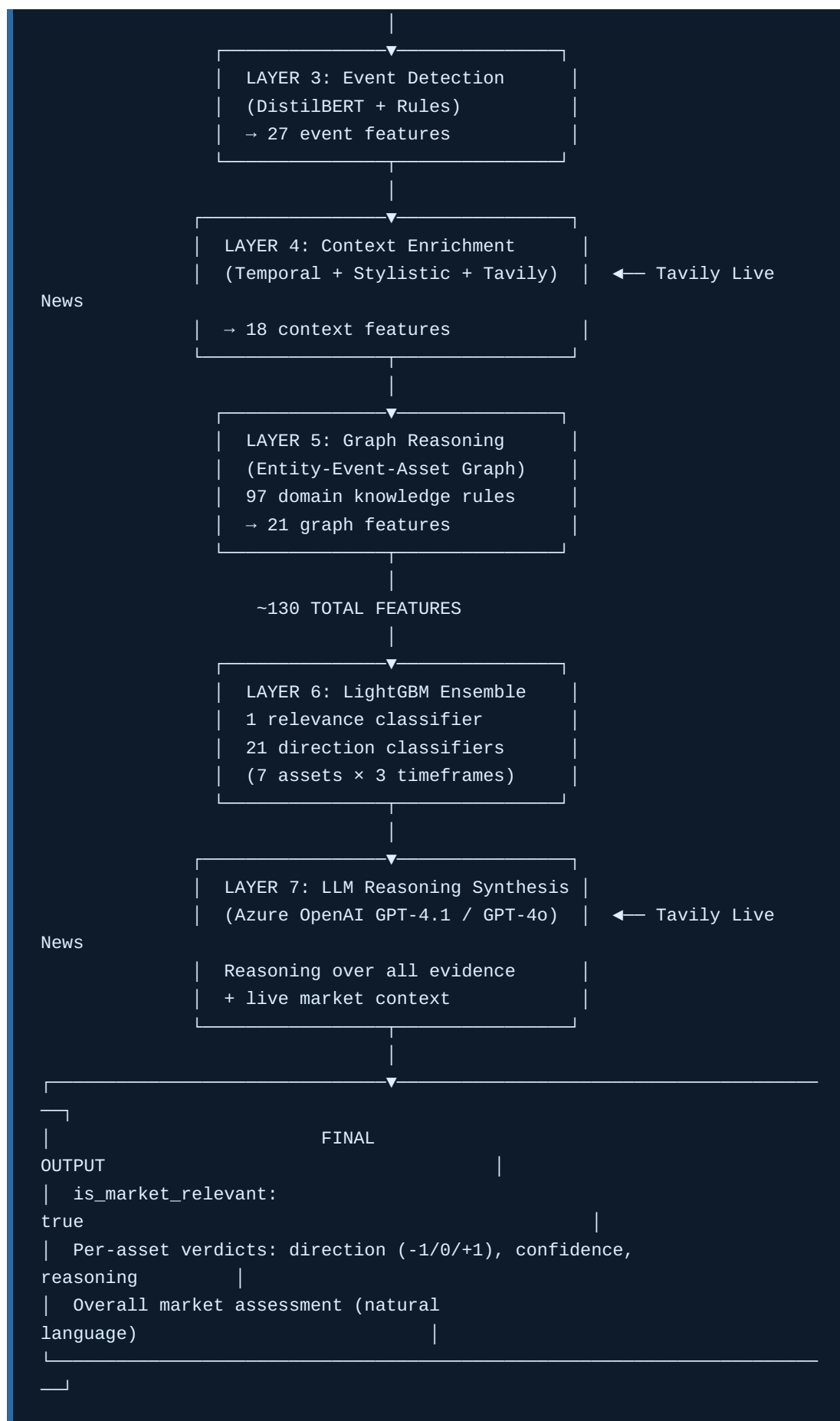
2. Running Example

- > **Tweet:** "TARIFFS on China! 150% immediately!"
- > **Posted:** During regular US market hours (10:30 AM ET)

This single tweet is used throughout every layer below to show exactly how the system processes it from raw text to final trading signal.

3. System Architecture





4. Layer-by-Layer Walkthrough

Layer 1 — Named Entity Recognition (GLiNER)

Purpose: Extract the financially and politically relevant entities from raw tweet text.

How it works:

- Uses **GLiNER** (a zero-shot NER model, ~170M parameters) to identify 10 entity types: persons, countries, organizations, commodities, policy terms, financial instruments, and more.
- Each detected entity is mapped to one or more of the seven target assets via a **curated lookup table** (e.g., "China" → equities, crude oil, wheat).
- Falls back to keyword matching when a GPU is unavailable.

Input → Output:

Input	Output
Raw tweet text	14 binary and count features per entity type

Example — Running Tweet:

```
Tweet: "TARIFFS on China! 150% immediately!"
```

```
Entities Detected:
```

```
- "China" → type: country → maps to: equities, crude oil, wheat
```

- "TARIFFS" → type: policy → maps to: equities, eurodollar
- "150%" → type: magnitude → signals: high urgency / severity

Entity Features (sample):

```
has_country: 1
has_policy_term: 1
entity_equities_signal: 0.85
entity_cl_signal: 0.70
entity_wheat_signal: 0.60
```

Feeds into: Layer 3 (event detection) and Layer 5 (graph reasoning).

Layer 2 — Financial Sentiment Analysis (FinBERT)

Purpose: Classify the emotional tone of the tweet specifically for financial language.

How it works:

- Uses **FinBERT**, pre-trained on financial news corpora (not general text). This makes it significantly more calibrated for market-relevant language than a general BERT model.
- Produces three probabilities: positive (bullish), negative (bearish), neutral — plus a compound score in $[-1, 1]$.

Input → Output:

Input	Output
Tweet text	8 sentiment features (probabilities, compound score, confidence)

Example — Running Tweet:

```
Tweet: "TARIFFS on China! 150% immediately!"
```

FinBERT Output:

```
positive_prob: 0.08 (not optimistic)
negative_prob: 0.81 (strong bearish tone)
neutral_prob: 0.11
compound_score: -0.74 (strongly negative)
```

→ Market interpretation: aggressive, bearish-toned announcement

Feeds into: Layer 5, where compound score modulates per-asset signal strength by $\pm 30\%$.

Layer 3 — Event Detection (DistilBERT + Trainable Head)

Purpose: Classify what type of geopolitical or economic event the tweet describes.

How it works:

- A **frozen DistilBERT encoder** (~66M parameters) generates a tweet embedding.
- A lightweight **trainable 2-layer MLP head** (~26K parameters) classifies the tweet into 13 event categories.
- A **hybrid overlay of 100+ regex patterns** ensures rare but critical event types (e.g., military conflict, crypto sanctions) are caught even when they appear too infrequently in training data to learn statistically.
- Multi-label: a tweet can trigger multiple event types simultaneously.

13 Event Categories:

Trade war / Tariffs · Sanctions · Military conflict · Monetary policy · Fiscal policy · Election · Regulation · Energy crisis · Pandemic · Central bank action · Currency volatility · Inflation signals · Structural reform

Input → Output:

Input	Output
Tweet text	27 features (one-hot per event type + confidence score per event)

Example — Running Tweet:

```
Tweet: "TARIFFS on China! 150% immediately!"

Events Detected:
  trade_war:      0.94  ✓ (primary event)
  sanctions:      0.38  (partial – tariffs are a form of trade
restriction)
  fiscal_policy:  0.31  (secondary signal)

Regex overlay confirms: keyword "TARIFFS" matches trade_war pattern
```

Feeds into: Layer 5 (graph reasoning) and Layer 7 (LLM prompt context).

Layer 4 — Context Enrichment

Purpose: Enrich the feature set with temporal context, tweet stylistic signals, and live macro news at prediction time.

How it works — three components:

Temporal Features

- Trading session: pre-market, regular hours (09:30–16:00 ET), post-market
- Hour of day, day of week, time since market open

Stylistic Features

- ALL CAPS ratio (urgency indicator)
- Exclamation count, tweet length, URL presence
- Thread position (1st vs. later in a rapid-fire series)
- Inter-tweet timing (seconds since previous tweet from same account)

Live Market News (Tavily Search API)

- Builds a keyword query from tweet text + timestamp and fetches the top 5 financial news snippets contemporaneous with the tweet
- Extracts binary macro flags: tariffs active, conflict ongoing, rate policy in focus, etc.
- These flags allow the statistical models to condition on whether macro conditions confirm the tweet's signal

Input → Output:

Input	Output
Tweet text + timestamp + live Tavily news	18 context features

Example — Running Tweet:

```
Tweet: "TARIFFS on China! 150% immediately!"
Timestamp: 2026-03-16 10:30 ET

Temporal:
  trading_session: "regular_hours" → market is open, immediate impact expected
  hour_of_day: 10 → active trading window

Stylistic:
  caps_ratio: 0.64 → HIGH urgency (TARIFFS, China)
  exclamation_count: 2 → aggressive announcement style

Tavily Live News (fetched):
  Headline 1: "US-China trade tensions escalate amid new tariff threats"
  Headline 2: "S&P 500 futures drop 1.2% on trade war fears"
  Macro flags: tariff_context=1, conflict_context=0, rate_context=0
```

Feeds into: Layer 6 (as numerical features) and Layer 7 (verbatim news snippets injected into LLM prompt).

Layer 5 — Entity-Event-Asset Graph Reasoning

Purpose: Encode structured financial domain knowledge about how political events propagate to specific asset classes.

How it works:

- Builds a **directed weighted graph** per tweet with three node types: **entities → events → assets**
- 97 hand-crafted **transmission rules** encode market wisdom (e.g., textbook economics, analyst knowledge):
 - (country:china, trade_war) → equities: (direction: -1, weight: 0.80)
 - (event:trade_war) → gold: (direction: +1, weight: 0.75) — safe-haven demand
 - (event:rate_hike) → treasury_2y: (direction: +1, weight: 0.85)
- The **FinBERT compound score** from Layer 2 modulates each signal by $\pm 30\%$ depending on bearish or bullish tone
- Graph topology metrics (density, centrality) become additional features

Why rules instead of purely learned? With only 2,669 labeled tweets, a model cannot learn rare but important event combinations. Rules generalize to unseen scenarios — for example, a conflict involving a country not present in training data.

Input → Output:

Input	Output
Entities (L1) + events (L3) + sentiment (L2)	21 features (per-asset signal direction/magnitude + graph topology)

Example — Running Tweet:

Graph constructed for "TARIFFS on China! 150% immediately!":

```
graph TD
    china[china (entity)] -- triggers --> trade_war[trade_war (event)]
    trade_war --> equities[equities (-1, 0.80)]
    trade_war --> gold[gold (+1, 0.75)]
    trade_war --> wheat[wheat (-1, 0.55)]
    equities --> cl[cl (-1, 0.70)]
    wheat --> eurodollar[eurodollar (-1, 0.65)]
```

FinBERT modulation (compound: -0.74):
→ Each signal strength increased ~22% (strong negative tone confirms bearish)

Final graph signals:

equities:	-0.98	(strong bearish)
gold:	+0.93	(strong bullish – safe haven)
cl:	-0.86	(bearish – demand destruction fears)
wheat:	-0.67	(bearish – trade disruption)
eurodollar:	-0.80	(bearish – USD strengthens vs EUR)
treasury_2y:	+0.75	(bullish – flight to safety)
btc:	+0.10	(weak signal – crypto less directly connected)

Feeds into: Layer 6 (21 graph features in the full ~130 feature vector) and Layer 7 (per-asset signals summarized in LLM prompt).

Layer 6 — LightGBM Statistical Ensemble

Purpose: Apply trained statistical classifiers to the ~130 assembled features to produce calibrated probabilistic predictions.

How it works:

- **22 independently trained LightGBM models:**
 - 1 binary relevance classifier (market-relevant vs. noise)
 - 21 direction classifiers: 7 assets × 3 timeframes (1m, 5m, 10m)
 - Each model predicts a direction $\in \{-1, 0, +1\}$ and outputs class probabilities
 - Trained on **2,669 historical tweets** with a **5-fold stratified cross-validation**
 - Class weights are inverse-frequency balanced to handle the skewed $\{-1, 0, +1\}$ distribution
 - **Separate models per asset:** each asset learns its own feature weights — sanctions news matters more for oil than for Bitcoin

Input → Output:

Input	Output
~130 features (Layers 1-5)	is_relevant (bool), relevance_score, per-asset direction + confidence

Example — Running Tweet:

```
Layer 6 Output (5m horizon):

Relevance:      RELEVANT (score: 0.92) ✓

Per-asset LightGBM predictions:
equities:      BEARISH  (-1, confidence: 0.81)
gold:          BULLISH  (+1, confidence: 0.74)
cl:            BEARISH  (-1, confidence: 0.69)
wheat:         BEARISH  (-1, confidence: 0.62)
eurodollar:    BEARISH  (-1, confidence: 0.76)
treasury_2y:   BULLISH  (+1, confidence: 0.68)
btc:           NEUTRAL  (0,  confidence: 0.53)
```

Feeds into: Layer 7 — all predictions passed as a statistical prior to the LLM.

Layer 7 — LLM Reasoning Synthesis (Azure OpenAI GPT-4.1 / GPT-4o)

Purpose: Final intelligence layer. Reasons over all structured evidence plus live market context to produce calibrated, interpretable verdicts — and provide natural-language explanations.

How it works — three steps:

Step 1: Real-Time Context Retrieval (Tavily)

- Builds a timestamp-aware query and fetches top 10 financial news snippets contemporaneous with the tweet
- Grounds the LLM in current market conditions, not just its training-time knowledge

Step 2: Structured Prompt Assembly

The LLM receives a complete structured prompt containing:

- Tweet text
- Extracted entities (Layer 1)
- Sentiment scores and compound tone (Layer 2)
- Detected event types and confidence (Layer 3)
- Trading session and macro flags (Layer 4)
- Per-asset graph propagation signals (Layer 5)
- **LightGBM predictions per asset (Layer 6)** — used as a statistical prior
- Live Tavily news snippets — verbatim market context

Step 3: GPT-4.1 / GPT-4o Call

- `response_format: json_object` — ensures structured, parseable output

- Temperature: 0.1 — low temperature for consistent, conservative verdicts
- The LLM reasons over all evidence, can **confirm or override** LightGBM on a per-asset basis, and must justify each decision in natural language

Input → Output:

Input	Output
All layer outputs + live Tavily news	Per-asset: direction, confidence (0–1), reasoning text + overall market assessment

Example — Running Tweet:

Tweet: "TARIFFS on China! 150% immediately!"

LLM Overall Assessment:

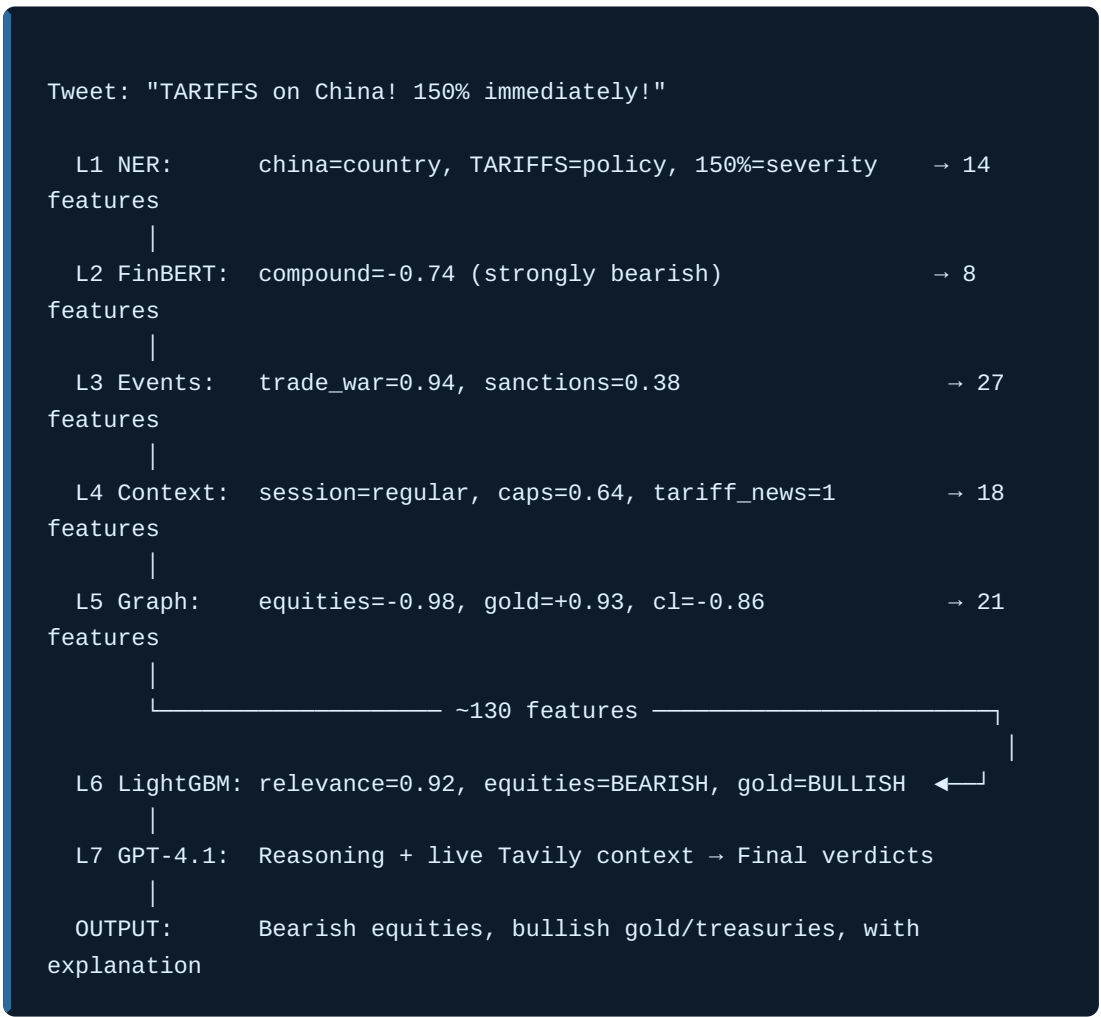
"Extreme trade war escalation signal. 150% tariffs on China represent a severe supply-chain shock. Equities face direct earnings risk. Gold and Treasuries benefit as safe havens. EUR/USD weakens as USD demand surges. Crude oil faces demand destruction from reduced global trade volumes."

Per-Asset LLM Verdicts (GPT-4.1, 5m horizon):

gold:	BULLISH	(0.82)	"Trade war escalation boosts safe-haven demand."
equities:	BEARISH	(0.89)	"150% tariffs severely damage corporate earnings and growth."
btc:	NEUTRAL	(0.51)	"Crypto shows limited sensitivity to trade war events."
cl:	BEARISH	(0.76)	"Trade war fears reduce global energy demand expectations."
wheat:	BEARISH	(0.63)	"China tariff retaliation may hit US agricultural exports."
eurodollar:	BEARISH	(0.84)	"USD safe-haven demand strengthens against EUR."
treasury_2y:	BULLISH	(0.81)	"Flight to safety boosts short-term Treasury demand."

Agrees with LightGBM: 6/7 assets | Latency: ~4,400ms

5. Data Flow Summary



6. Key Design Decisions

Decision	Rationale
Modular 7-layer architecture	Each layer can be debugged, replaced, or improved independently without retraining the full system
No transformer fine-tuning	Only 2,669 training tweets — fine-tuning large transformers would overfit. Layers 1–3 use frozen or minimally trained models
Hybrid event detection (neural + rules)	Rules ensure rare but high-impact events (military conflict, sanctions) are always caught, even when underrepresented in training data
Graph transmission rules	97 domain rules encode textbook market knowledge and generalize to event combinations never seen in training
Separate LightGBM per asset	Each asset responds differently to the same event. Separate models prevent cross-asset interference
Tavily for real-time grounding	Without live news, both Layer 4 features and the LLM reason from stale context. Tavily synchronizes prediction with current market conditions
LightGBM as LLM prior	The LLM receives statistical predictions as input — it reasons against them and can confirm or override, with its reasoning visible
Temporal train/test split	The oldest 80% trains, the most recent 20% tests. Prevents look-ahead bias and mirrors real deployment conditions

7. Inference Latency Breakdown

Component	Latency
Layers 1–3 (NER, Sentiment, Events)	~120ms

Component	Latency
Layer 4 (Tavily feature call)	~200–800ms
Layer 5 (Graph reasoning)	~5ms
Layer 6 (LightGBM ensemble)	~5ms
Layer 7 (GPT-4.1 + Tavily context)	~3,500–5,000ms
Total end-to-end	~4,000–6,000ms

8. Evaluation Results

Evaluated on **667 held-out test samples** using a strict temporal split (no test data seen during training).

Relevance Classification Accuracy — Three Approaches

Model	Approach	Accuracy	Precision	Recall
GPT-4.1	LLM Reasoning (Layer 7 variant)	88.01%	82.29%	55.63%
LightGBM	Statistical Ensemble (Layer 6)	86.96%	75.23%	57.75%
GPT-4o	LLM Reasoning (Layer 7 baseline)	85.61%	77.38%	45.77%

Key insight: GPT-4.1 achieves the highest relevance accuracy and the highest precision — a 7-point precision advantage over LightGBM. Higher precision means fewer false signals sent to downstream trading decisions, directly

reducing costly whipsaws. At ~20,000 API calls/day, GPT-4.1 filtering costs approximately \$12–16/month — a low price for the quality improvement.

Direction Prediction Accuracy (5m horizon, mean across 7 assets)

Model	Mean Accuracy	Strength
LightGBM (Layer 6)	44.61%	Statistical pattern recognition
GPT-4.1 (Layer 7)	23.17%	Fundamentals-driven assets (Wheat, Treasury)
GPT-4o (Layer 7)	22.04%	Reasoning and interpretability

Note: LightGBM leads on raw directional accuracy across most assets. LLMs match or beat LightGBM on **Wheat (56% vs 53%)** and **2Y Treasury (38% vs 37%)** — precisely the assets most sensitive to geopolitical narrative and monetary policy, where context understanding outperforms pattern matching.

9. Recommended Production Configuration

The evaluation supports a **hybrid deployment** that leverages the strengths of both approaches:

Step 1 – GPT-4.1 relevance gate (88% accuracy, 82% precision)
→ Filter out noise. Only proceed if tweet is market-relevant.

Step 2 – LightGBM direction prediction (44.6% mean accuracy)

→ Fast, calibrated statistical direction on confirmed relevant signals.

Step 3 – GPT-4.1 reasoning synthesis

→ Final verdict with natural-language justification, confidence score,
and context grounding via Tavily.

Result: Fewer trades, higher confidence, with full interpretability — the correct optimization for profitable, risk-managed trading.

Report Generated: 2026-03-16 | System: General Market Impact Classifier |
Test Set: 667 samples (temporal split)